

Setting Up Arduino

Fall 2026

Setting up Arduino IDE

- We will be using the Arduino IDE (and writing “C++”)
- IDE stands for Integrated Development Environment.
 - Write your code
 - Compile your code
 - Download your code to the robot
- First, download Arduino IDE from this website (<https://tinyurl.com/arduinoDownload>) if you want to set it up on your own computer

Legacy Moment

Please download this version
(find it by scrolling down the Downloads page)

Legacy IDE (1.8.X)



Arduino IDE 1.8.19

The open-source Arduino Software (IDE) makes it easy to write code and upload it to the board. This software can be used with any Arduino board.

Refer to the [Arduino IDE 1.x documentation](#) for installation instructions.

SOURCE CODE

Active development of the Arduino software is [hosted by GitHub](#). See the instructions for [building the code](#). Latest release source code archives are available [here](#). The archives are PGP-signed so they can be verified using [this](#) gpg key.

DOWNLOAD OPTIONS

Windows Win 7 and newer

Windows ZIP file

Windows app Win 8.1 or 10



Linux 32 bits

Linux 64 bits

Linux ARM 32 bits

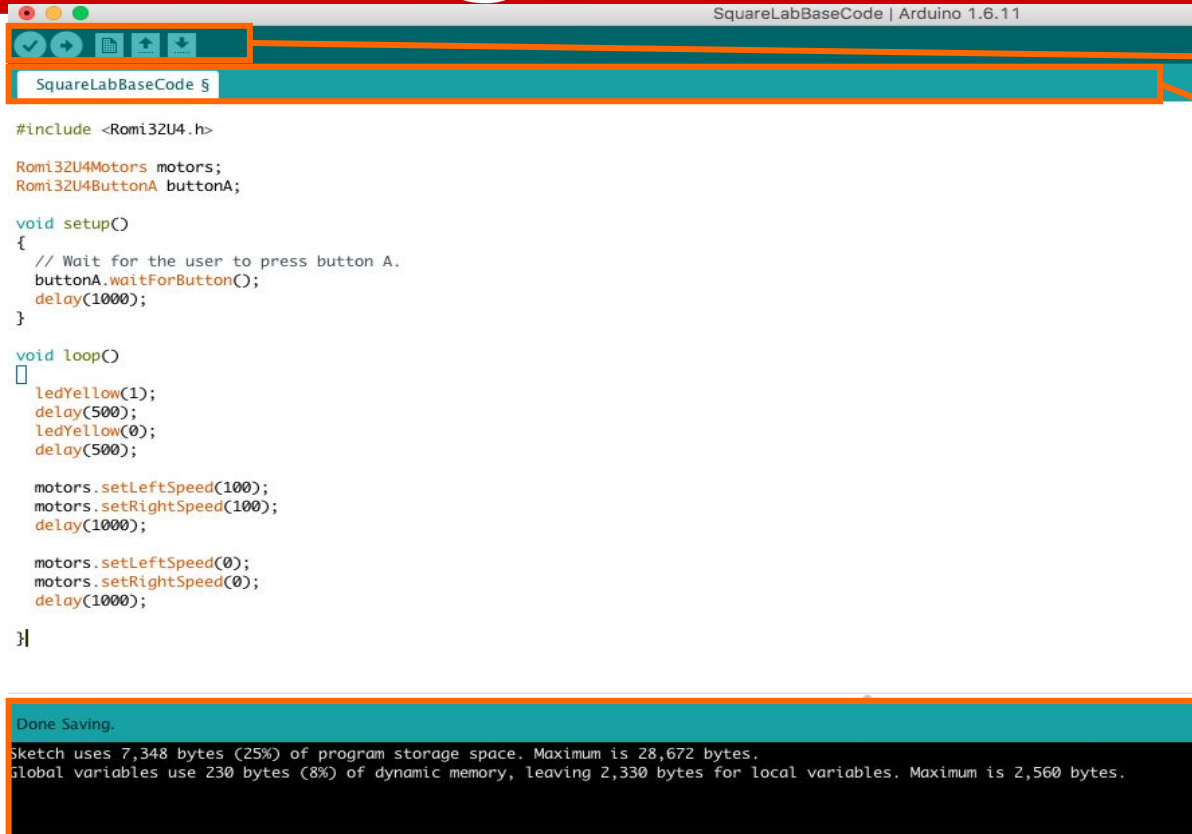
Linux ARM 64 bits

Mac OS X 10.10 or newer

[Release Notes](#)

[Checksums \(sha512\)](#)

Using Arduino



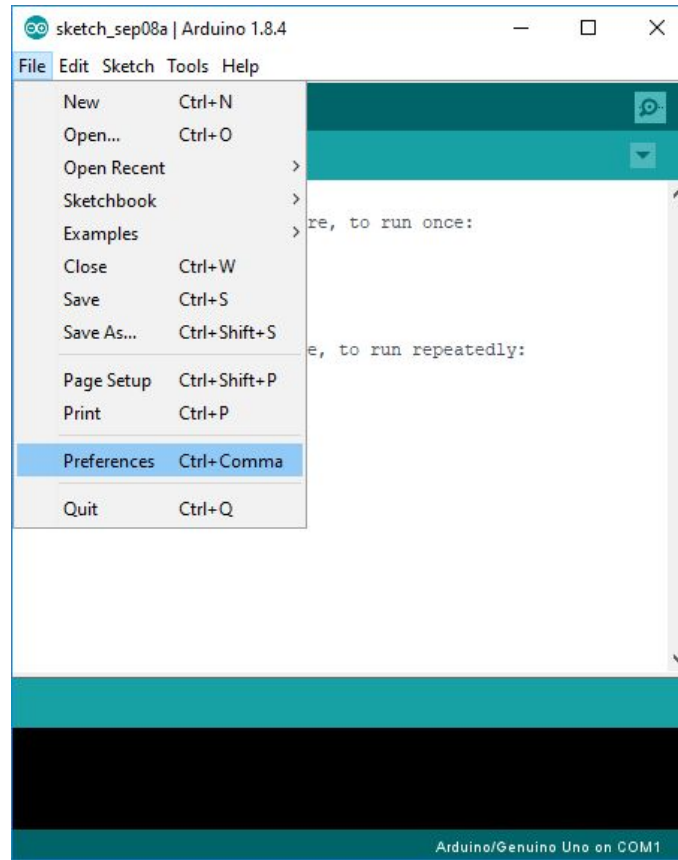
Tool bar

File tabs

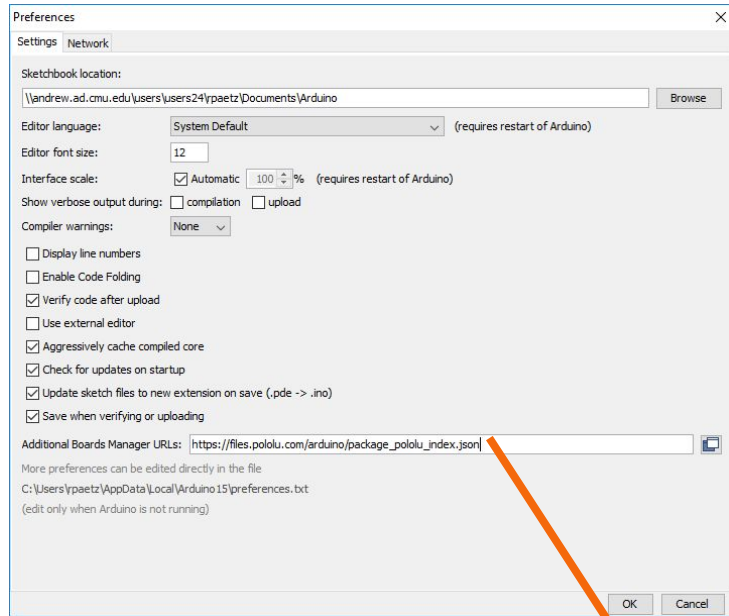
Current file

Output

Setting Up Arduino

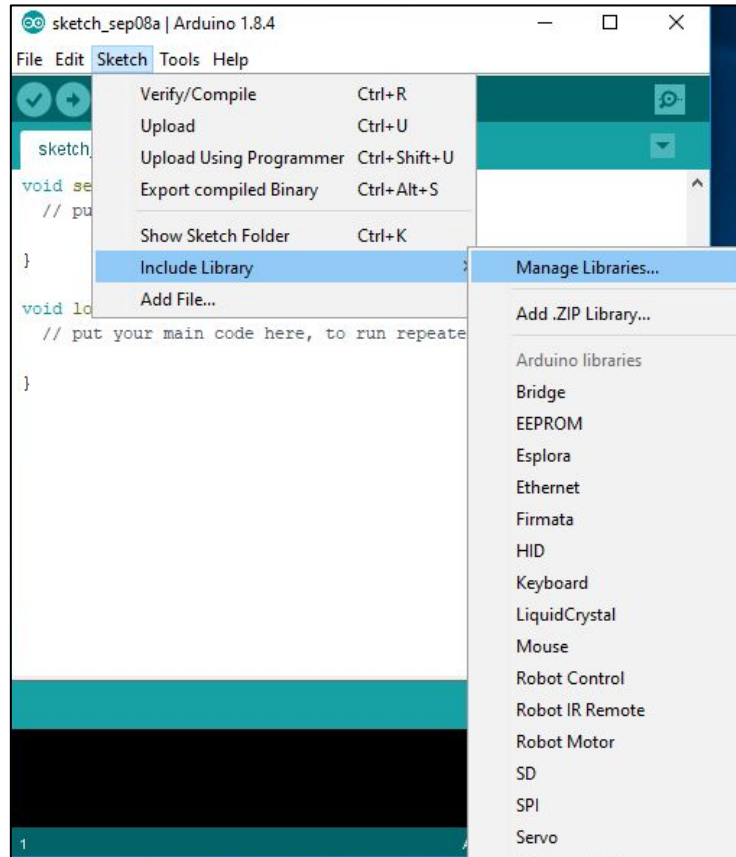


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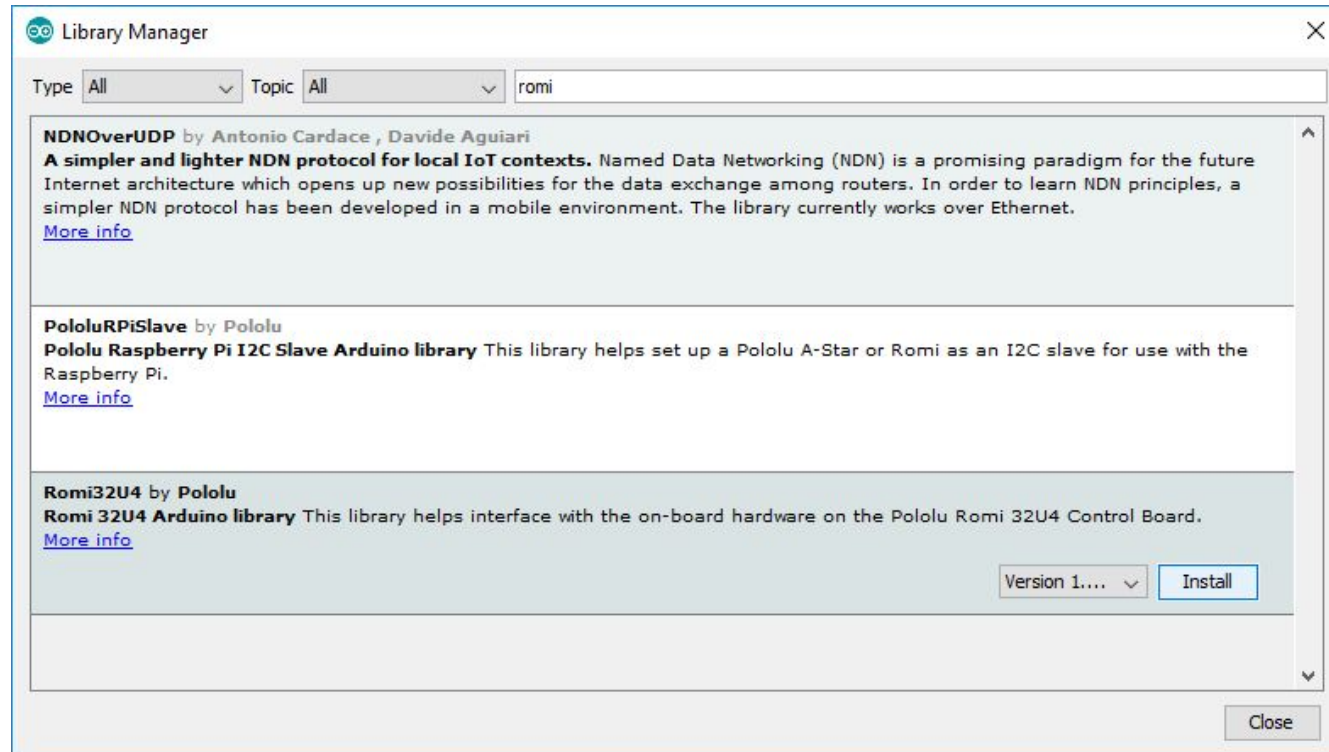


https://files.pololu.com/arduino/package_pololu_index.json

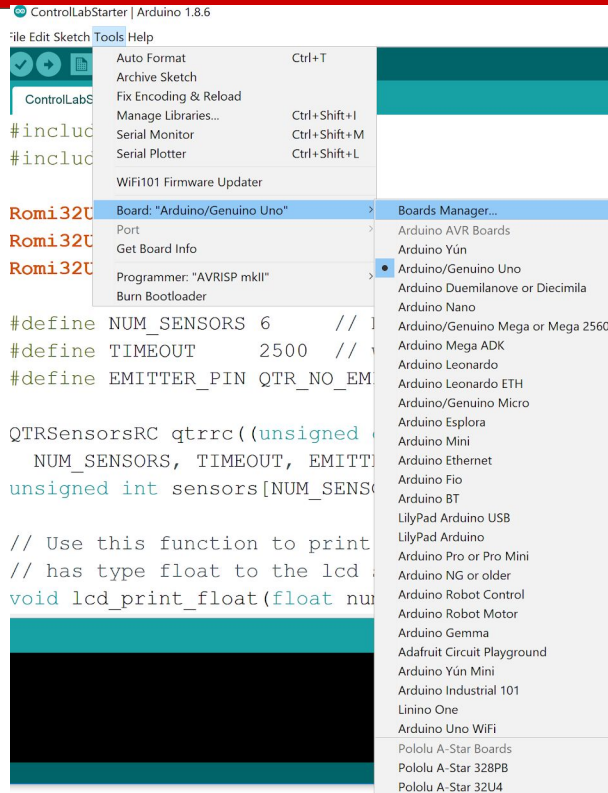
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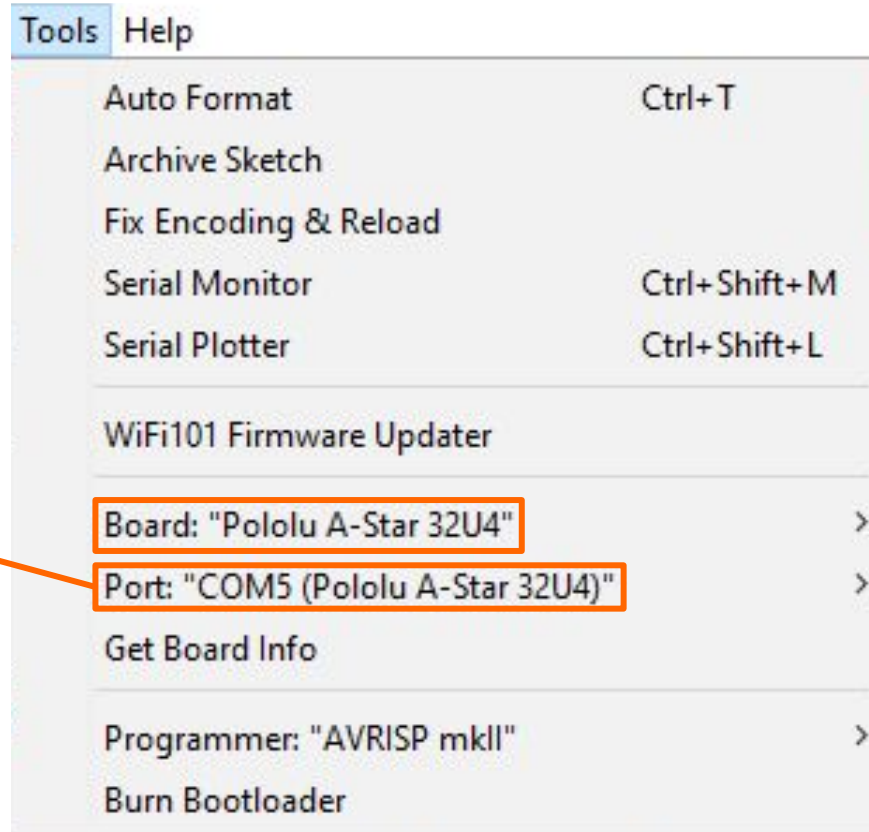
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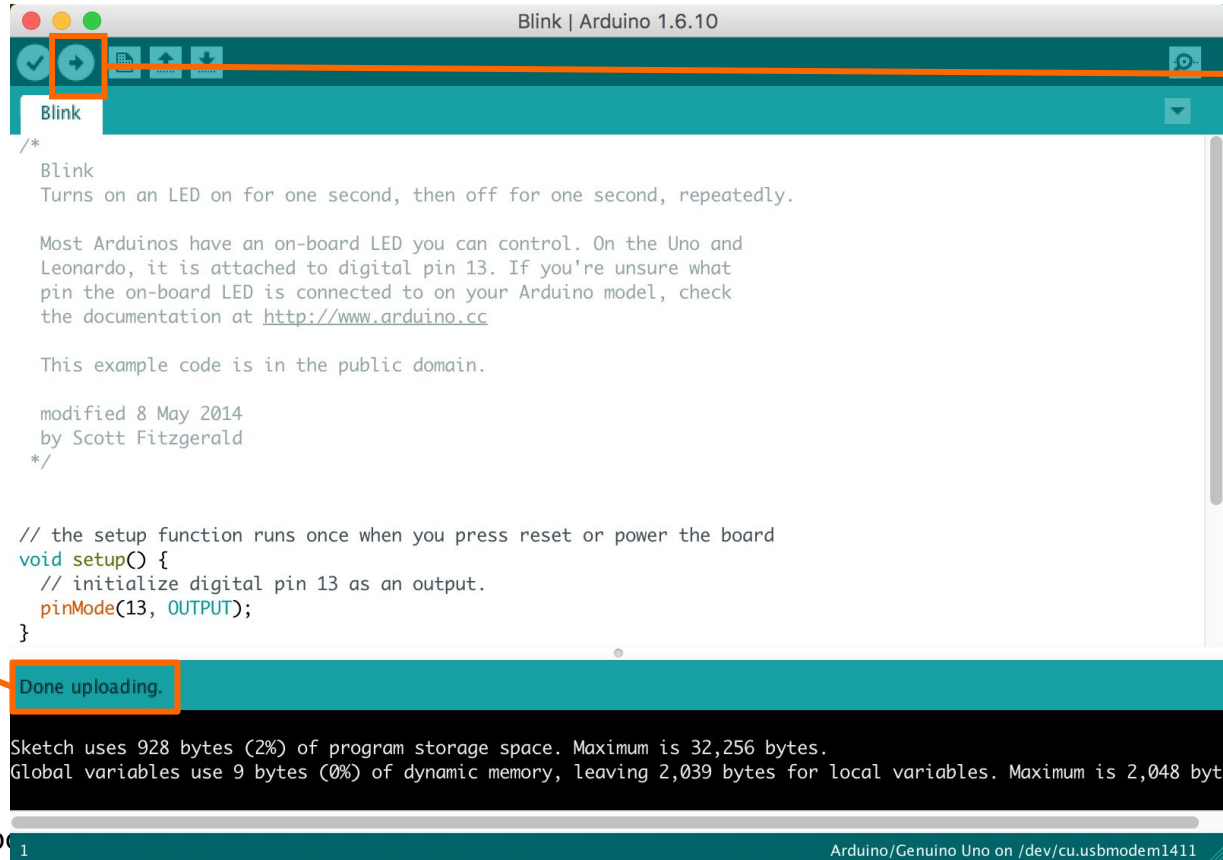


Selecting correct settings



Don't forget to
pick your
programmer's
port too!!

Uploading Code



Click!

All
Good!

Code Skeleton Breakdown

```
#include <Romi32U4.h>
```

Includes Libraries

```
Romi32U4Motors motors;  
Romi32U4ButtonA buttonA;
```

Creates Objects

```
void setup()  
{  
  // Wait for the user to press button A.  
  buttonA.waitForButton();  
  delay(1000);  
}
```

Set-Up Code
(Only Runs Once!)

```
void loop()  
{  
  ledYellow(1);  
  delay(500);  
  ledYellow(0);  
  delay(500);  
  
  motors.setLeftSpeed(100);  
  motors.setRightSpeed(100);  
  delay(1000);  
  
  motors.setLeftSpeed(0);  
  motors.setRightSpeed(0);  
  delay(1000);  
}
```

Main Code
(Loops Until Robot is
Turned Off)

Code Command Breakdown

```
// Run once, when the sketch starts
void setup() {
  uint16_t batteryLevel;

  // Loop the code within { } until A is pressed
  while (!buttonA.isPressed()) {
    // read the battery level and store it in a variable
    batteryLevel = readBatteryMillivolts();

    lcd.clear(); // clear the LCD
    lcd.print("B=");
    lcd.print(batteryLevel); // print the battery voltage
    lcd.print("mV");
    lcd.gotoXY(0, 1);
    lcd.print("Press A");

    delay(100); // Pause for 100ms
  }
  lcd.clear();

  delay(500);
}
```

Comments!
Allows programmer to
write notes without
influencing code

Waits to run the code
until button A is
pressed

LCD prints "Press A"

Code Command Breakdown

```
void loop()
```

```
{
```

```
  ledYellow(1);
```

```
  delay(500);
```

```
  ledYellow(0);
```

```
  delay(500);
```

```
  motors.setLeftSpeed(100);
```

```
  motors.setRightSpeed(100);
```

```
  delay(1000);
```

```
  motors.setLeftSpeed(0);
```

```
  motors.setRightSpeed(0);
```

```
  delay(1000);
```

```
}
```

Set left motor at
power level 100
(-200 to 200)

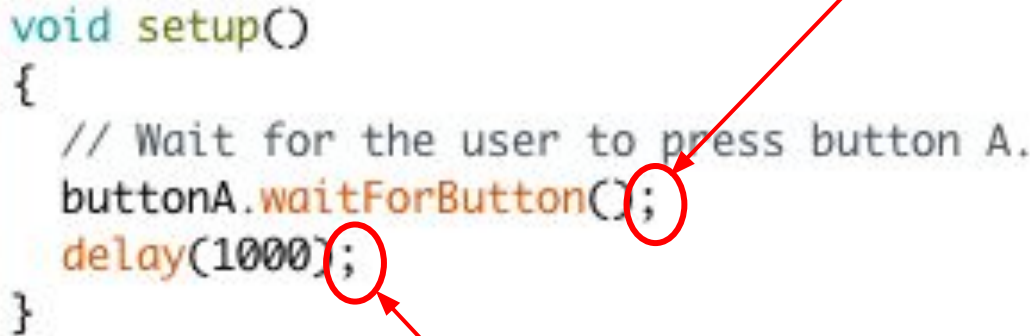
Set right motor at
power level 100
(-200 to 200)

Stop thinking (allow
current setting to
continue) for 1000 ms

Remember the Semicolon!

There must be a semicolon (;) at the end of every line!

```
void setup()
{
  // Wait for the user to press button A.
  buttonA.waitForButton();
  delay(1000);
}
```

A diagram illustrating the importance of semicolons in code. It shows a code snippet with two semicolons circled in red. Red arrows point from the top right towards each semicolon, emphasizing their presence at the end of the lines.